

Feature Engineering, Model Optimization & Performance Comparison

AI & Machine Learning - Task 2 Report

Abstract

This project compares Linear Regression, Ridge Regression and Decision Tree Regressor on the California Housing dataset.

Objective

Build and compare regression models using feature scaling and evaluation metrics.

Dataset

California Housing dataset from scikit-learn with housing-related features and median house value target.

Tools

Python, Pandas, NumPy, Matplotlib, Scikit-learn, VS Code.

Methodology

Loaded dataset, checked missing values, split train/test data, applied StandardScaler, trained three regression models and evaluated using RMSE and R^2 .

Results

Decision Tree Regressor achieved the best performance in this implementation (RMSE $\approx$ 0.7030, $R^2 \approx$ 0.6228).

Conclusion

The Decision Tree model outperformed Linear Regression and Ridge Regression for this task. Further tuning could improve accuracy.

Model	RMSE	R^2
Linear Regression	0.7456	0.5758
Ridge Regression	0.7456	0.5758
Decision Tree	0.7030	0.6228